

Expt 7

Spectral Estimation

7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

Objectives:

1. To estimate the spectrum of a signal using different window functions such as Rectangular, Hamming, and Hanning windows.
2. To analyze the effect of window selection on spectral leakage.
3. To compare the main-lobe width and side-lobe levels of different windows.
4. To evaluate the accuracy of frequency component detection.

Procedure:

1. Open MATLAB and create a new script file.
2. Define the sampling frequency and number of samples.
3. Generate a discrete-time signal consisting of sinusoidal frequency components.
4. Apply the Rectangular window to the signal.
5. Compute the FFT and estimate the spectrum.
6. Repeat the process using Hamming and Hanning windows.
7. Plot the spectral estimates obtained using each window.
8. Compare the spectra in terms of spectral leakage, main-lobe width, and side-lobe levels.
9. Observe the visibility of frequency components for each window.

Program (Matlab Code):

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Signal
```

```
fs = 1000;
```

```
N = 256;
```

```
t = (0:N-1)/fs;
```

```

x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);

NFFT = 2048;
f = (0:NFFT/2-1)*(fs/NFFT);

%% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular', 'Hamming', 'Hanning'};

%% Processing
for i = 1:3

    xw = x(:) .* w{i};
    X = fft(xw, NFFT);

    P{i} = abs(X(1:NFFT/2));
    P{i} = P{i} / max(P{i});

end

%% Individual Spectrums
figure;

for i = 1:3

    subplot(3,1,i);

    plot(f, 20*log10(P{i}), 'LineWidth', 1.5);

    title(['Spectrum using ' names{i} ' Window']);
    xlabel('Frequency (Hz)');

```

```

        ylabel('Magnitude (dB)');

        xlim([0 250]);

        grid on;
    end

%% Comparison
figure;

plot(f, 20*log10(P{1}), 'k', ...
     f, 20*log10(P{2}), 'r', ...
     f, 20*log10(P{3}), 'b', ...
     'LineWidth', 1.5);

legend(names);
title('Window Comparison');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xlim([50 170]);
grid on;

%% Window Shapes
figure;

for i = 1:3
    subplot(3,1,i);

    plot(w{i}, 'LineWidth', 1.5);
    title([names{i} ' Window']);
    xlabel('Sample');
    ylabel('Amplitude');
end

```

```

    grid on;
end

%% Peak Detection
for i = 1:3

    [pks{i}, loc{i}] = findpeaks(P{i}, f, ...
        'MinPeakHeight', 0.3);

    fprintf('\n%s Window Frequencies:\n', names{i});
    disp(loc{i});

end

%% Peak Marking (Hamming)
figure;

plot(f, 20*log10(P{2}), 'LineWidth', 1.5);
hold on;

plot(loc{2}, 20*log10(pks{2}), ...
    'ro', 'MarkerFaceColor', 'r');

title('Detected Frequencies (Hamming)');
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
xlim([50 170]);
grid on;

%% Main Peaks

```

```
fprintf('\nMain Spectral Peaks:\n');
```

```
for i = 1:3
```

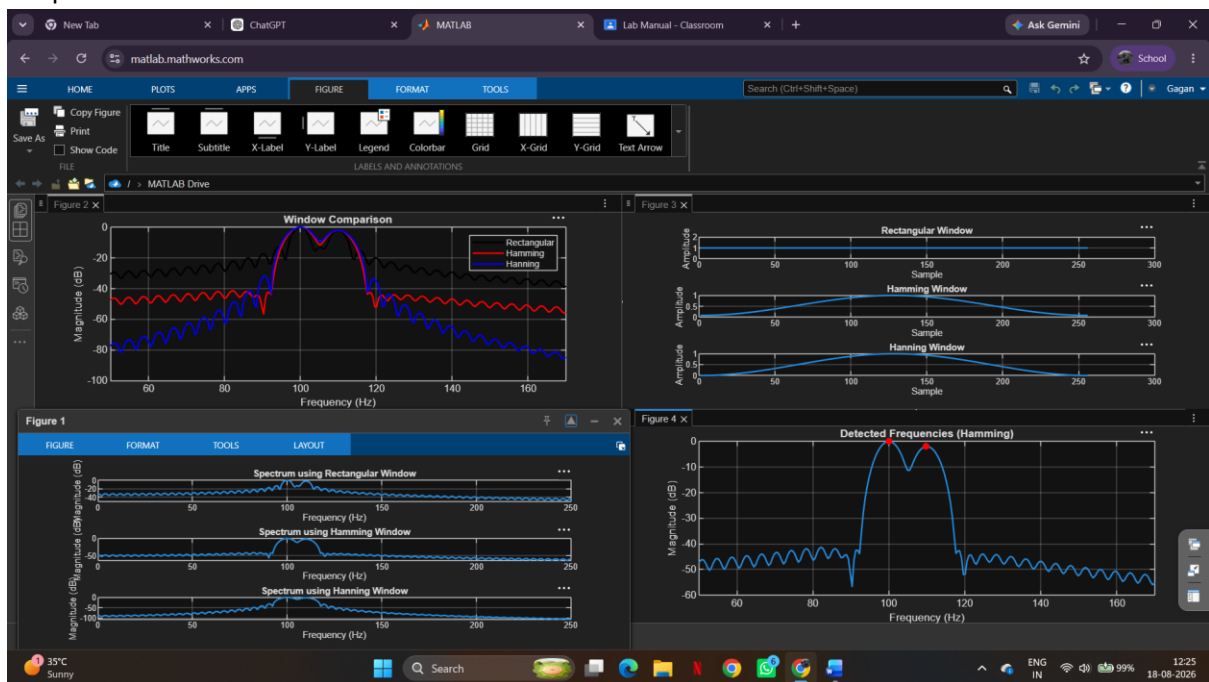
```
    [~, idx] = max(P{i});
```

```
    fprintf('%s Window Peak = %.2f Hz\n', ...
```

```
        names{i}, f(idx));
```

```
end
```

Output:



Rectangular Window Frequencies:

100.0977 109.8633

Hamming Window Frequencies:

100.0977 109.8633

Hanning Window Frequencies:

100.0977 109.8633

Main Spectral Peaks:

Rectangular Window Peak = 100.10 Hz

Hamming Window Peak = 100.10 Hz

Hanning Window Peak = 100.10 Hz

>>

Spectrum Estimation

Output

Detected Frequencies using Rectangular Window (Hz)

100.0977 109.8633

Detected Frequencies using Hamming Window (Hz)

100.0977 109.8633

Detected Frequencies using Hanning Window (Hz)

100.0977 109.8633

Result:

1. The spectrum of the given signal was successfully estimated using Rectangular, Hamming, and Hanning window functions in MATLAB.
2. The effect of window selection on spectral leakage was analyzed, showing that the Rectangular window exhibits higher leakage compared to Hamming and Hanning windows.
3. The main-lobe width and side-lobe levels were compared, where Hamming and Hanning windows provided better side-lobe suppression at the cost of slightly wider main-lobes.
4. The accuracy of frequency component detection was evaluated, and it was observed that windowing improves frequency resolution and reduces spectral distortion.